```
=== SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION (Go-Back-N) ===

Enter total number of frames to send : 6
Enter sender's window size : 4
Enter frame number to simulate as LOST
(0 to 5, enter -1 for no loss): 2

------------------------------------------
Sender Window : [Frame 0 - Frame 3]
Node A --> Sending Frame 0
Node A --> Sending Frame 1
Node A --> Sending Frame 2
   (Frame 2 LOST in transit!)
Node A --> Sending Frame 3
Node B --> Frame 2 missing. Discarding out-of-order frames after it.
Node A --> Timeout! Resending window starting from Frame 2.

------------------------------------------
Sender Window : [Frame 0 - Frame 3]
Node A --> Sending Frame 2
Node A --> Sending Frame 3
Node B --> Received Frame 0, sending ACK 0
Node B --> Received Frame 1, sending ACK 1
Node B --> Received Frame 2, sending ACK 2
Node B --> Received Frame 3, sending ACK 3
Node A --> ACK 3 received. Sliding window forward.

------------------------------------------
Sender Window : [Frame 4 - Frame 5]
Node A --> Sending Frame 4
Node A --> Sending Frame 5
Node B --> Received Frame 4, sending ACK 4
Node B --> Received Frame 5, sending ACK 5
Node A --> ACK 5 received. Sliding window forward.

------------------------------------------
All 6 frames successfully transmitted from Node A to Node B.
```

EXNO:7 FLOW CONTROL AT DATA LINK LAYER USING SLIDING WINDOW PROTOCOL

Aim:

To implement the sliding window for flow control and simulate frame transmission between two nodes.

Algorithm:

1) Start the program.
2) Read the total number of frames, window size, and lost frame.
3) Initialize base = 0 and next_seq = 0.
4) Send frames within the current window.
5) If a frame is lost, retransmit from the lost frame
6) Receive Ack and slide the window forward
7) Repeat until all frames are transmitted.
8) Stop.

Program:

```
def simulate (n, w, lost):
  base = next_seq = 0
  while base <n :
    print (f" In window : {base} to {min
          (base +w-1 , n-1)}")
while next_seq < base + w and next_seq<n :
  print (" send Frame", next_seq)
if next_seq == lost :
  print (" Frame lost !")
```

```
next_seq += 1
if lost is not None and base <= lost < next_seq:
    print("Timeout! Resending from frame", lost)
    next_seq = lost
    lost = None
    continue
for i in range(next_seq):
    base = next_seq
print("\n All frames transmitted successfully.")
n = int(input("Frames: "))
w = int(input("Window size: "))
lost = int(input("Lost frame (-1 for none): "))
simulate(n, w, None if lost == -1 else lost)
```

Result:

The sliding window protocol was implemented successfully and all frames were transmitted with proper flow control and error recovery.